MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIFTH SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCCSC301 - Operating Systems

(2024 ADMISSION ONWARDS)

MODEL QUESTION PAPER

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each Course Outcome (CO) to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer All Questions
Each question carries 2 marks

1. Define an Operating System. [U] [CO1]
2. List any two types of operating systems, giving an example of each. [K] [CO1]
3. Name any two CPU scheduling criteria. [U] [CO2]
4. What is a deadlock? [U] [CO3]
5. State the concept of virtual memory. [U] [CO4]

(5 x 2 = 10 Marks)

Part B
Short Answer Questions
Answer any 6 Questions
Each question carries 5 marks

6. Distinguish between different types of user interfaces provided by an operating system. [U] [CO1]
7. Discuss the purpose of system calls and how a system call is made. [U] [CO1]
8. List the different types of system calls, giving one example of each. [K] [CO1]
9. Outline the contents of a Process Control Block (PCB). [U] [CO2]
10. Distinguish between long-term, short-term, and medium-term schedulers. [U] [CO2]
11. Describe the wait() and signal() operations of a semaphore, and their role in process synchronization. [U] [CO3]
12. Discuss the methods used for recovery from deadlock. [U] [CO3]
13. Illustrate the FIFO page replacement algorithm with a suitable example. [A] [CO4]

(6 x 5 = 30 Marks)

Part C
Essay Questions
Answer any 2 Questions
Each question carries 15 marks

14. Discuss the various methods of Inter-process Communication (IPC), with suitable examples of shared memory and message-passing systems. [A] [CO2]
15. Analyse the working of the Banker's Algorithm for deadlock avoidance with a suitable example. Compare its effectiveness with deadlock detection and recovery techniques in ensuring safe resource allocation. [An] [CO3]
16. Illustrate the memory management strategies of Contiguous Memory Allocation, Paging, and Segmentation, with suitable diagrams. [A] [CO4]

(2 x 15 = 30 Marks)